

1 • Postulates and measurement

- State: normalized $|\psi\rangle = \sum c_n |a_n\rangle$, $c_n = \langle a_n | \psi \rangle$, $\sum |c_n|^2 = 1$. Global phase unphysical; relative phases physical.
- Outcomes = eigenvalues of Hermitian $\hat{A}$. Born: $P(a_n) = |\langle a_n | \psi \rangle|^2$. Degenerate: add over the eigenspace.
- Collapse: after a_n the state is $|a_n\rangle$ (degenerate: project onto the eigenspace, renormalize). Repeat at once $\rightarrow$ same result. Measuring prepares states; N -dim space $\Rightarrow \leq N$ outcomes.
- $\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle = \sum a_n |c_n|^2$ (ensemble average, need not be an eigenvalue). $\sigma_A^2 = \langle A^2 \rangle - \langle A \rangle^2$; $\sigma_A = 0 \Leftrightarrow$ eigenstate.
- $\langle \phi | \psi \rangle$ (bar) = inner product; $\langle A \rangle$ (no bar) = average.
- Superposition: add amplitudes, then square (cross term $2\text{Re}(c_1^* c_2 \dots)$). Mixture: add probabilities. Only observables not commuting with the prep basis tell them apart.

2 • Dirac notation and linear algebra

- Ket = column, bra = conjugate-transposed row. $\langle \phi | \psi \rangle = \sum \phi_n^* \psi_n = \int \phi^* \psi dx$; $\langle \psi | \phi \rangle = \langle \phi | \psi \rangle^*$.
- Orthonormal $\langle e_m | e_n \rangle = \delta_{mn}$; completeness $\sum |e_n\rangle \langle e_n| = \hat{1}$; $A_{mn} = \langle e_m | \hat{A} | e_n \rangle$ (row m , col n).
- $A^\dagger = (A^T)^*$; $(AB)^\dagger = B^\dagger A^\dagger$, $(AB)^{-1} = B^{-1} A^{-1}$, $(|a\rangle \langle b|)^\dagger = |b\rangle \langle a|$.
- Hermitian: real eigenvalues, orthogonal eigenvectors, complete basis, real averages. AB Hermitian $\Leftrightarrow [A, B] = 0$; $i[A, B]$ always Hermitian.
- Unitary $U^\dagger U = 1$: keeps lengths; $|\lambda| = 1$; products unitary, sums not. $e^{i\hat{A}}$ unitary for Hermitian $\hat{A}$.
- Spectral: $\hat{A} = \sum a_n |a_n\rangle \langle a_n|$, $f(\hat{A}) = \sum f(a_n) |a_n\rangle \langle a_n|$. Projector $P = \sum |e\rangle \langle e|$; $P^2 = P$, eigenvalues 0, 1.
- 2×2 : $\lambda = \frac{1}{2} (\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det})$; $\text{tr} = \Sigma \lambda$, $\det = \Pi \lambda$. Rows dependent $\Rightarrow \det 0 \Rightarrow$ no inverse.
- Gram-Schmidt: $|e'_k\rangle \propto |e_k\rangle - \sum_{j < k} \langle e'_j | e_k \rangle |e'_j\rangle$ (conjugate on the bra). Cauchy-Schwarz $|\langle v | w \rangle|^2 \leq \langle v | v \rangle \langle w | w \rangle$.
- $[A, BC] = [A, B]C + B[A, C]$. $[\hat{x}, \hat{p}] = i\hbar$, $[\hat{x}, \hat{p}^2] = 2i\hbar\hat{p}$, $[f(\hat{x}), \hat{p}] = i\hbar f'(\hat{x})$, $[\hat{H}, \hat{x}] = -\frac{i\hbar}{m}\hat{p}$.
- Continuous: $\langle x | x' \rangle = \delta(x - x')$, $\Psi(x) = \langle x | \psi \rangle$, $\langle x | p \rangle = \frac{e^{ipx/\hbar}}{\sqrt{2\pi\hbar}}$, $\Phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx/\hbar} \Psi(x) dx$. $\hat{p} \rightarrow -i\hbar \partial_x$ (x -space), $\hat{x} \rightarrow i\hbar \partial_p$ (p -space).

3 • Wavefunctions

- $\int |\Psi|^2 dx = 1$ ($|\Psi\rangle = L^{-1/2}$). $\langle f(x) \rangle = \int f |\Psi|^2$; $\langle p \rangle = \int \Psi^* (-i\hbar \partial_x) \Psi$; $\langle p^2 \rangle = \hbar^2 \int |\Psi'|^2$ (safe at kinks).
- **Shortcuts**: $|\Psi|^2$ symmetric about $x_0 \Rightarrow \langle x \rangle = x_0$; real Ψ ($\times$ const phase) $\Rightarrow \langle p \rangle = 0$; factor $e^{ik_0 x}$ adds $\hbar k_0$ to $\langle p \rangle$; time phases drop out of $|\Psi|^2$.
- Current $J = \frac{\hbar}{m} \text{Im}(\Psi^* \Psi')$; $\partial_t |\Psi|^2 + \partial_x J = 0$; $A e^{ikx}$: $J = |A|^2 \hbar k / m$; real Ψ : $J = 0$.
- $\frac{d}{dt} \langle Q \rangle = \frac{i}{\hbar} \langle [\hat{H}, \hat{Q}] \rangle + \langle \partial_t \hat{Q} \rangle$. Ehrenfest: $\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}$, $\frac{d\langle p \rangle}{dt} = -\langle V' \rangle$ (exact Newton only if $V \leq$ quadratic).

Exam moves: see this $\rightarrow$ do this

- $P(E_n)$, $\Psi(x, t)$, $\langle H \rangle \rightarrow$ recipe; symmetry kills c_n ; trig-polynomial Ψ : expand by identity; $\langle H \rangle$ via $\int |\Psi'|^2$.
- **Sudden change** $\rightarrow \Psi$ unchanged; overlaps with the new ψ'_n .
- **Free packet** $\rightarrow$ Fourier; shift rules; complete the square; σ_p fixed.
- **Wells, steps** $\rightarrow$ match ψ , ψ' , divide; count branches; T as flux.
- **Matrix observable** $\rightarrow$ eigen by $\text{tr}/\det$; Born with the conjugated eigenvector; degenerate $\rightarrow$ project.
- **Prove an identity** $\rightarrow$ insert $1 = \sum |n\rangle \langle n|$; or $\frac{d}{dt} \langle Q \rangle$ by commutator.
- **Qubit** $\rightarrow$ Bloch vector; $\hat{H} = \vec{a} \cdot \vec{\sigma}$ is an axis; time = rotation.
- **Always** $\rightarrow$ strategy first (points); units, $\sum P = 1$, limits.

Integrals and identities

- $\int_0^\infty x^n e^{-\alpha x} dx = \frac{n!}{\alpha^{n+1}}$. $\int e^{-\alpha x^2 - \beta x} dx = \sqrt{\frac{\pi}{\alpha}} e^{\beta^2/4\alpha}$.
- $\int x^2 e^{-\alpha x^2} = \frac{1}{2\alpha} \sqrt{\frac{\pi}{\alpha}}$
- $\int_{-\infty}^\infty \frac{du}{(1+u^2)^2} = \int \frac{u^2 du}{(1+u^2)^2} = \frac{\pi}{2}$; antiderivative $\frac{1}{2} \left[\frac{u}{1+u^2} + \arctan u \right]$
- $\int_0^a \sin^2 \frac{n\pi x}{a} = \frac{a}{2} - \int_0^a x \sin^2 = \frac{a^2}{4} - \int_0^a x^2 \sin^2 = a^3 \left(\frac{1}{6} - \frac{1}{4n^2\pi^2} \right)$
- $\int_0^a x(a-x) \sin \frac{n\pi x}{a} dx = \frac{2a^3}{(n\pi)^3} [1 - (-1)^n]$.
- $\int_0^a x \psi_1 \psi_2 dx = -\frac{16a}{9\pi^2}$
- $\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$. $\sin^3 u = \frac{3}{4} \sin u - \frac{1}{4} \sin 3u$. $\sin u \cos^2 u = \frac{1}{4} (\sin u + \sin 3u)$
- $|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2 + 2\text{Re}(z_1^* z_2) = \cos \theta + i \sin \theta \cdot A e^{ikx} + B e^{-ikx} = (A+B) \cos kx + i(A-B) \sin kx$

4 • Stationary states and the recipe

- $\Psi = \psi(x) e^{-iEt/\hbar}$, $\hat{H}\psi = E\psi$, $\hat{H} = -\frac{\hbar^2}{2m} \partial_x^2 + V$. Stationary: $|\Psi|^2$ and all $\langle Q \rangle$ frozen, $\sigma_H = 0$.
- **Recipe**: (1) solve $\hat{H}\psi_n = E_n \psi_n$; (2) $c_n = \int \psi_n^* \Psi(x, 0) dx$; (3) $\Psi(x, t) = \sum c_n \psi_n e^{-iE_n t/\hbar}$. $P(E_n) = |c_n|^2$, $\langle H \rangle = \sum |c_n|^2 E_n$: **constant** for any state.
- $\langle H \rangle$ shortcut: $\frac{\hbar^2}{2m} \int |\Psi'|^2 + \int V |\Psi|^2$. Jumps in $\Psi \Rightarrow \langle H \rangle = \infty$; kinks $\Rightarrow$ finite $\langle H \rangle$ but maybe $\sigma_H = \infty$.
- Two energies $\Rightarrow$ density beats at $\omega = (E_m - E_n)/\hbar$. Phase $e^{i\varphi}$ on c_2 shifts $\cos(\omega t) \rightarrow \cos(\omega t - \varphi)$.
- ψ continuous; ψ' continuous unless $V = \infty$ (or δ). $E > V_{\min}$. Bound states: real, non-degenerate (1-D), n -th has $n-1$ nodes; even $V \Rightarrow$ even/odd, ground state even.
- $\psi'' = \frac{2m}{\hbar^2} (V - E) \psi$: $E > V$ oscillates (bigger $E - V \Rightarrow$ shorter λ , smaller amplitude); $E < V$ decays.
- Inverse: $V(x) = E + \frac{\hbar^2}{2m} \frac{\psi''}{\psi}$; use the node-free ground state. $V(x_0) = E$ where $\psi'' = 0$.

5 • Infinite square well $[0, a]$

- $\psi_n = \sqrt{\frac{2}{a}} \sin \frac{n\pi x}{a}$, $E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} = n^2 E_1$, $E_{n+1} - E_n = (2n+1)E_1$.
- $\langle x \rangle = \frac{a}{2}$, $\langle x^2 \rangle = a^2 \left(\frac{1}{3} - \frac{1}{2n^2 \pi^2} \right)$, $\langle p^2 \rangle = \left(\frac{n\pi \hbar}{a} \right)^2$, $\sigma_x \sigma_p = \frac{\hbar}{2} \sqrt{\frac{n^2 \pi^2}{3} - 2}$ (0.568 $\hbar$ at $n=1$).
- Centered $[-\frac{a}{2}, \frac{a}{2}]$: cos for odd n , sin for even; same E_n .
- **Symmetry**: Ψ symmetric about $a/2 \Rightarrow$ only odd n ; antisymmetric $\Rightarrow$ only even n .
- $\frac{1}{\sqrt{2}} (\psi_1 + \psi_2)$: $\langle x \rangle = \frac{a}{2} - \frac{16a}{9\pi^2} \cos \omega t$, $\omega = \frac{3\pi^2 \hbar}{2ma^2}$. Revival $T = \frac{4ma^2}{\hbar \pi^2}$.
- Sudden change: Ψ unchanged, expand in new states; $\langle H \rangle$ unchanged. Ground state, wall $a \rightarrow 2a$: $P(E'_1) = \frac{32}{9\pi^2}$, $P(E'_2) = \frac{1}{2}$; both walls out symmetrically: $\frac{64}{9\pi^2}$.
- Known c_n : $x(a-x)$: $\frac{8\sqrt{15}}{(n\pi)^3}$ (odd); triangle: $\frac{4\sqrt{6}}{(n\pi)^2} \sin \frac{n\pi}{2}$; left half flat: $\frac{2}{n\pi} (1 - \cos \frac{n\pi}{2})$.

6 • Free particle, Fourier, δ

- e^{ikx} , $E = \frac{\hbar^2 k^2}{2m}$: not normalizable. $\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int \phi(k) e^{i(kx - \hbar k^2 t/2m)} dk$, $\phi(k) = \frac{1}{\sqrt{2\pi}} \int \Psi(x, 0) e^{-ikx} dx$; $\int |\phi|^2 = \int |\Psi|^2$.
- $v_g = \frac{d\omega}{dk} = \frac{\hbar k}{m}$ (particle), $v_{ph} = \frac{v_g}{2}$. σ_p constant; packet spreads at σ_p/m .
- Gaussian $(\frac{2a}{\pi})^{1/4} e^{-ax^2}$: $\sigma_x = \frac{1}{2\sqrt{a}}$, $\sigma_p = \hbar \sqrt{a}$ (minimum), $\langle H \rangle = \frac{\hbar^2 a}{2m} + \frac{\hbar^2 k_0^2}{2m}$ if $e^{ik_0 x}$; $\sigma(t) = \sigma_0 \sqrt{1 + (\frac{\hbar t}{2m\sigma_0^2})^2}$; $w = \sqrt{a}/\sqrt{1 + (2\hbar a t/m)^2}$, $|\Psi|^2 = \sqrt{\frac{2}{\pi}} w e^{-2w^2 x^2}$.
- Pairs: Gaussian $\leftrightarrow$ Gaussian; box half-width $L \leftrightarrow \frac{\sin kL}{k}$ (zeros $k = n\pi/L$); $\sqrt{k} e^{-\kappa|x|} \leftrightarrow \sqrt{\frac{\kappa}{2\pi}} \frac{2\kappa}{\kappa^2 + k^2} \cdot f e^{ik_0 x} \leftrightarrow F(k - k_0)$; $f(x - x_0) \leftrightarrow e^{-ikx_0} F$. Narrow $\leftrightarrow$ wide.
- $\int f \delta(x - a) = f(a)$; $\delta(cx) = \frac{\delta(x)}{|c|}$; $\frac{1}{2\pi} \int e^{ik(x-x')} dk = \delta(x - x')$; $(|x|)'' = 2\delta(x)$.

7 • Finite well $(-V_0, |x| < a)$ and scattering

- $\ell = \frac{\sqrt{2m(E+V_0)}}{\hbar}$, $\kappa = \frac{\sqrt{-2mE}}{\hbar}$, $z = \ell a$, $z_0 = \frac{\sqrt{2mV_0}}{\hbar}$, $E = -V_0 + \frac{\hbar^2 z_0^2}{2ma^2}$.
- Even: $\kappa = \ell \tan \ell a \Rightarrow \tan z = \sqrt{z_0^2/z^2 - 1}$. Odd: $\kappa = -\ell \cot \ell a \Rightarrow -\cot z = \sqrt{z_0^2/z^2 - 1}$ (needs $z_0 > \frac{\pi}{2}$).
- $N = \lceil 2z_0/\pi \rceil$; always ≥ 1 even state. Deep: $E_n + V_0 \approx \frac{n^2 \pi^2 \hbar^2}{2m(2a)^2}$ (slightly lower: leakage). $P_{\text{out}} = \frac{\cos^2 z / \kappa a}{1 + \frac{\sin^2 z}{\kappa^2 a} + \cos^2 z / \kappa a}$.
- Penetration $1/\kappa$; electron, 1 eV below rim: 0.195 nm.
- Well, $E > 0$: $T^{-1} = 1 + \frac{V_0}{4E(E+V_0)} \sin^2(2\ell a)$; $T = 1$ when $2\ell a = n\pi$.
- Step ($E > V_0$) or drop: $R = \left(\frac{k-k'}{k+k'} \right)^2$, $T = \frac{4kk'}{(k+k')^2} = \frac{k'}{k} |t|^2$ (flux). $E < V_0$ step: $R = 1$, evanescent tail, $J = 0$.
- Barrier $E < V_0$: $T^{-1} = 1 + \frac{V_0^2 \sinh^2 \kappa L}{4E(V_0 - E)}$, $\kappa L \gg 1$: $T \approx \frac{16E(V_0 - E)}{V_0^2} e^{-2\kappa L}$. $E > V_0$: $\sinh \rightarrow \sin$, $\kappa \rightarrow k'$; $T = 1$ at $k' L = n\pi$. Compare L with $1/\kappa$ (or $1/k'$).

8 • Uncertainty

- $\sigma_A^2 \sigma_B^2 \geq \left(\frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right)^2$ (uses the *average* commutator: can vanish). $\sigma_x \sigma_p \geq \frac{\hbar}{2}$ always. Commuting $\Rightarrow$ common eigenbasis.
- Equality: Gaussians, at minimum width. Spreads of the state, not measurement kicks.
- $\sigma_H \Delta t \geq \frac{\hbar}{2}$, $\Delta t = \sigma_Q / |d\langle Q \rangle/dt|$; time is not an operator. $\frac{1}{\sqrt{2}} (\psi_1 + \psi_2)$: $\sigma_H = \frac{E_2 - E_1}{2}$.
- Estimate: $\langle \hat{H} \rangle \gtrsim \frac{\hbar^2}{8m\sigma^2} + \langle V \rangle$, minimize. HO $\rightarrow \frac{1}{2} \hbar \omega$ (exact); H atom $\rightarrow a_0$, -13.6 eV.
- $\sigma_x \sigma_H \geq \frac{\hbar}{2m} |\langle p \rangle|$. Free: $\sigma_x^2(t) = \sigma_x^2(0) + \frac{\sigma_p^2 t^2}{m^2}$ (no initial $x-p$ correlation).

9 • Qubit / spin-1/2

- $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$: Hermitian, unitary, traceless, $\sigma^2 = 1$, eigenvalues ± 1 . $\vec{S} = \frac{\hbar}{2} \vec{\sigma}$.
- $\sigma_j \sigma_k = \delta_{jk} + i \epsilon_{jkl} \sigma_l$; $[\sigma_j, \sigma_k] = 2i \epsilon_{jkl} \sigma_l$; $(\vec{a} \cdot \vec{\sigma})(\vec{b} \cdot \vec{\sigma}) = \vec{a} \cdot \vec{b} + i(\vec{a} \times \vec{b}) \cdot \vec{\sigma}$.
- $|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\varphi} \sin \frac{\theta}{2} |1\rangle \leftrightarrow \hat{n}$; $(\vec{\sigma}) = \hat{n}$; orthogonal = antipodal. $|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}$, $|\pm y\rangle = \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}}$.
- $P(+\hat{m}) = \frac{1}{2} (1 + \hat{n} \cdot \hat{m}) = \cos^2 \frac{\Delta}{2}$. $\Delta \sigma_j = \sqrt{1 - \langle \sigma_j \rangle^2}$.
- $H = a_0 + \vec{a} \cdot \vec{\sigma}$: $E = a_0 \pm |\vec{a}|$, eigenstates along $\pm \vec{a}$. $e^{i\alpha \hat{n} \cdot \vec{\sigma}} = \cos \alpha + i \sin \alpha \hat{n} \cdot \vec{\sigma}$; 2π rotation = -1.
- $\vec{U} = e^{-i\hat{H}t/\hbar}$. $\hat{H} = \frac{\hbar}{2} \vec{\Omega} \cdot \vec{\sigma}$ precesses $\hat{n}$ about $\vec{\Omega}$ at rate $|\vec{\Omega}|$. From $|0\rangle$ with $\hbar \omega (\alpha \sigma_z + \beta \sigma_x)$: $P_1 = \frac{\beta^2}{R^2} \sin^2(\omega R t)$, $R = \sqrt{\alpha^2 + \beta^2}$. $\frac{\hbar \Omega}{2} \sigma_x$: $P_1 = \sin^2 \frac{\Omega t}{2}$. Ramsey: $P_1 = \cos^2 \frac{\omega T}{2}$.

Traps

- $|z|^2 = z^* z$, not z^2 . Conjugate when forming a bra.
- Normalize first; keep $\sqrt{2/a}$ in c_n ; each E_n its own phase.
- Cusp: $\langle p^2 \rangle$ from $\int |\Psi'|^2$, not Ψ'' (δ at the kink).
- Bloch angles are doubled: $\cos^2(\gamma/2)$, antipodal = orthogonal.
- T is a flux ratio (k'/k); thick-barrier formula needs $\kappa L \gg 1$. A "probability" with units $\Rightarrow$ non-normalizable state.

Constants

$\hbar = 1.055 \times 10^{-34}$ J·s = 6.582 $\times 10^{-16}$ eV·s · $\hbar c = 1240$ eV·nm · $\hbar c = 197.3$ eV·nm · $m_e = 9.109 \times 10^{-31}$ kg = 0.511 MeV/c² · $m_p/m_e = 1836$ · $\frac{\hbar^2}{2m_e} = 0.0381$ eV·nm² · $\lambda_e = \frac{1.226 \text{ nm}}{\sqrt{K[\text{eV}]}}$ · 1 eV = 1.602 $\times 10^{-19}$ J · electron in 1 nm box: $E_1 = 0.376$ eV